Jerry Xie

linkedin.com/in/chenxie-aa8266135 | cxiejobs@gmail.com | github.com/cxiesde-art | +1 (412) 482-7842

WORKING EXPERIENCE

Snorkel AI Sep 2025 – Present

Part-time Contractor (Python + Docker + Agent Framework + LoRA SFT)

- Contributing to Terminus Bench, an open-source coding agent evaluation benchmark providing sandboxed environments for safely executing and testing AI-generated code.
- Designed and implemented new coding agent eval benchmarks to expand task coverage and improve assessment reliability across diverse agent frameworks.
- Fine-tuned DeepSeek V4 Flash using LoRA SFT on curated domain-specific instruction datasets; implemented the end-to-end training pipeline with ms-swift framework, managed data preprocessing and tokenisation, and integrated the fine-tuned model into the benchmark evaluation pipeline to assess performance gains on coding agent tasks.
- **CyberSWE Security Benchmarking Project:** Developed benchmark tasks to evaluate AI coding agents on realistic vulnerability remediation. Seeded application-layer flaws in real repositories, wrote symptom-only task prompts, implemented hidden blackbox security tests, built deterministic Docker environments, and validated gold fixes through Harbor/Realm oracle workflows.

Weibo Oct 2024 – Jan 2026

Senior LLM Engineer (Python + Pytorch + Verl + RAG)

- Implemented a multi-agent report generation system like DeepResearch, enabling multi-turn retrieval and report generation.
- Using RAG as the knowledge source to build an Elon Musk-like AI agent with both short-term and long-term memory capabilities with Qwen model.
- Led the backend architecture for a multi-turn dialogue system. For the Follow-up Question module, built a user profiling-based precision recommendation algorithm. Leveraged high-quality conversational data for supervised fine-tuning (SFT) and applied reinforcement learning (RL)-based optimisation (DPO), resulting in significant improvements in dialogue quality and substantial increases in transition rate and click-through rate (CTR).
- Applied a Search-R1-inspired RL framework to optimize a custom search system, enabling the model to dynamically generate search queries and iteratively refine answers; integrated retrieved results into reasoning loops to improve accuracy and relevance of responses in complex, multi-step queries.
- Designed structured reasoning outputs to guide the model's search and decision-making process, enhancing both precision and interpretability.
- Implemented a modular pipeline supporting multiple LLMs and search backends, achieving significant improvement in retrieval efficiency and answer quality over baseline prompt-only approaches.

Microsoft Dec 2022 – Oct 2024

Software Engineer (Azure + Python + Pytorch)

- Implemented llama3, GPT-4o, Mistral and other base model fine-tuning pipeline on Azure ML studio. Managed training data on Azure blob container. Generated pipeline SBOM files (data tokenised, preprocessing, verification, prompt instructions, LoRA fine-tuning, deployment, evaluation).
- Implemented English and 10 multi-language (Chinese, Korean, etc.) training data pipelines for GPT-3.5 turbo, GPT-4 text summarization and fine-tuning evaluation.
- Used GPT-4 for data augmentation on small datasets to fine-tune GPT-3.5 turbo, LLaMA 3 models in English/multi-language. Evaluated fine-tuned models on Microsoft Teams meeting summaries.

Kafang Tech Sep 2019 – May 2021

Software Engineer (Go/C++ + Microservice + Kubernetes + gRPC + Vue.js)

- Designed and implemented ML training cloud-native platform. Deployed across two Kubernetes clusters. Built user authentication.
- Integrated Redis caching, reducing data retrieval time by 60% and improving app performance.
- Built 10+ RESTful APIs for training data management (upload, validation).
- Implemented Redis task queue to handle training tasks and decouple sidecar/gateway; built gRPC connection between gateway and queue.
- **Server monitoring and alerting:** Implemented dynamic memory alert rules in Prometheus (Golang, Python). Reduced false alerts from 46.6% to 17.6%. Configured email/Slack alerts in Alertmanager; visualized metrics with Grafana.
- **Mail notification:** Used go-mall package to create email notification module for user account management.

OPEN SOURCE PROJECTS

Terminus Bench | Coding Agent Evaluation Benchmark

github.com/laudeinstitute/terminal-bench — Python + Docker + Agent Framework

- Built a coding agent evaluation benchmark with sandboxed Docker environments for safely executing and testing AI-generated code across diverse agent frameworks.

EDUCATION

Carnegie Mellon University – Pittsburgh, PA Jun 2021 – Dec 2022

Master of Science in Computer Engineering (Software Engineering track)

Dalian University of Technology – Dalian, China Sep 2015 – Sep 2019

Bachelor of Engineering in Chemical Engineering (Computation track)